Edited v.s. Non-edited Regions

3

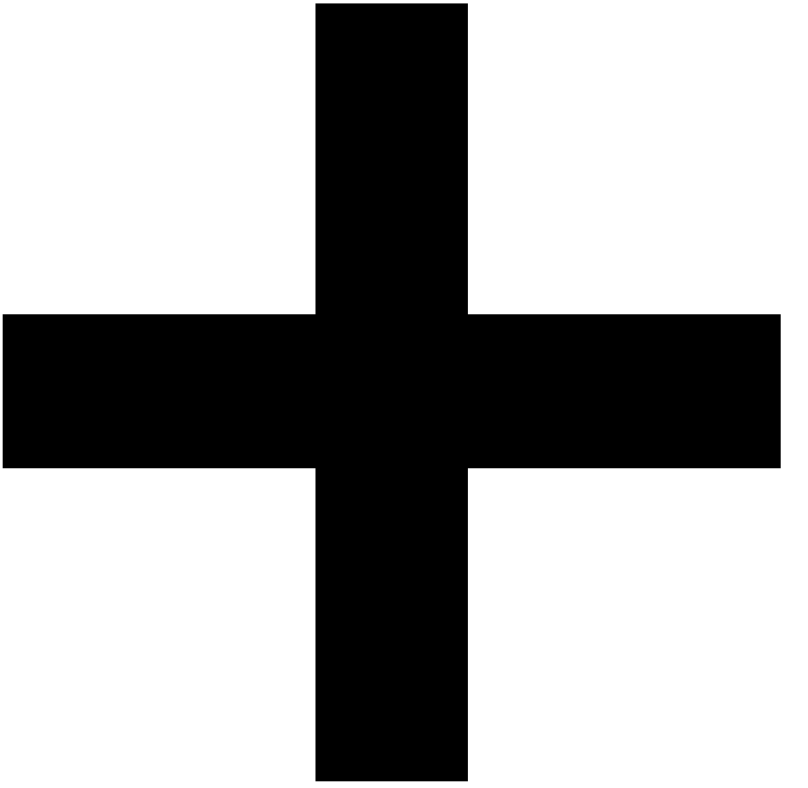
Change the tiger to a cat

in the segmented area

(a) a straightforward way

Two key goals:

1. **Fidelity / reconstruction**: ***non-edited area*** looks the same as the source video
2. **Instruction following**: ***the edited area*** aligns well with the user intent (~~**tiger**~~ **a cat**)

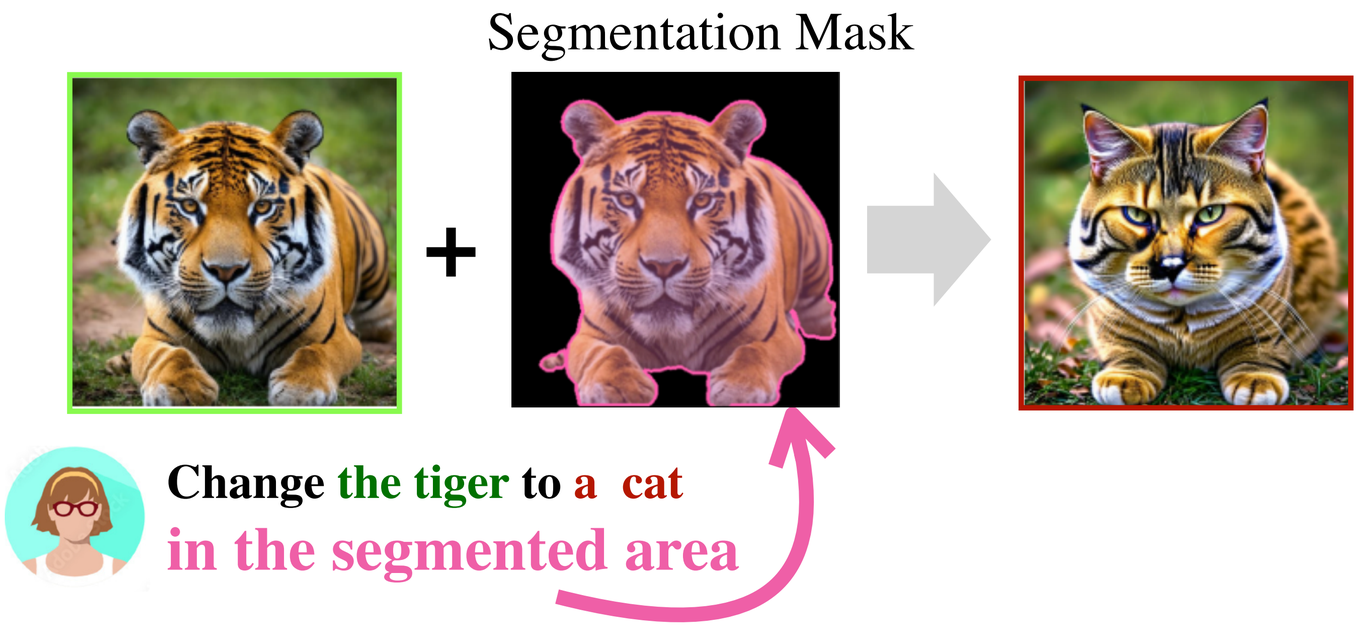


Segmentation Mask

Edited v.s. Non-edited Regions

Two key goals:

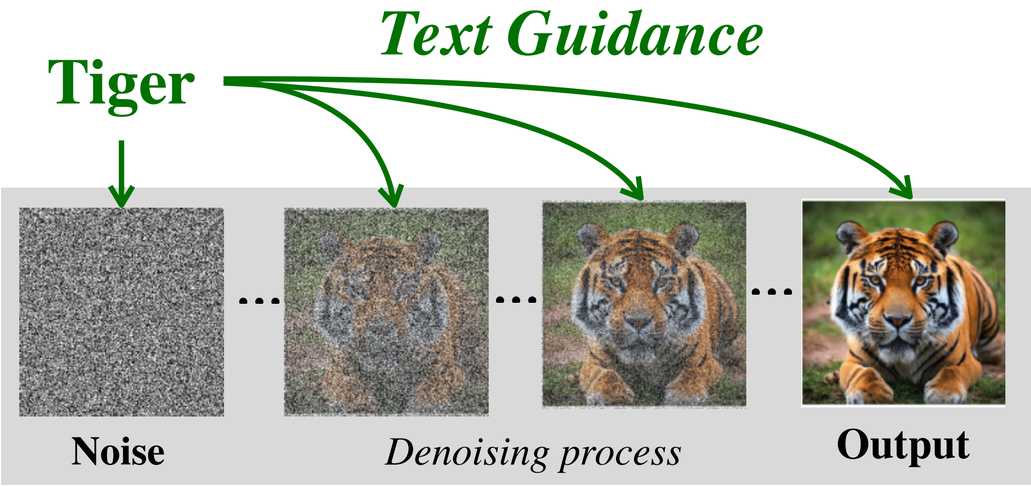
1. **Fidelity / reconstruction**: ***non-edited area*** looks the same as the source video
2. **Instruction following**: ***the edited area*** aligns well with the user intent (~~**tiger**~~ **a cat**)



(a) a straightforward way

Text-to-Image/Video Generation Models

Text-to-image generation



Implicitly learn text-image correspondence

Diffusion Models
(Stable Diffusion / Flux)

Text-to-video generation

Text prompt: *A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick.*

Video Diffusion Models
(Sora2)